

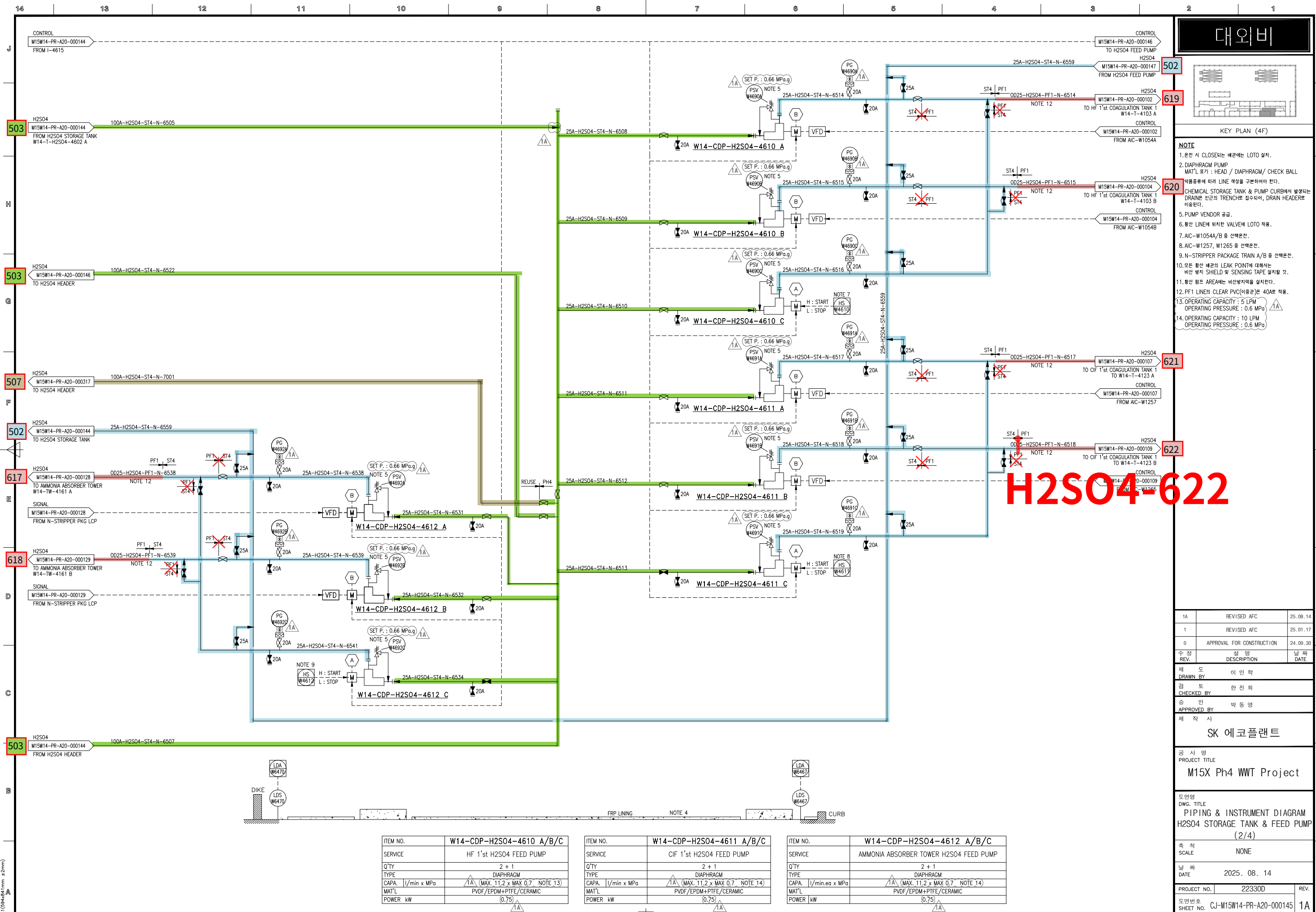
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

H2SO4-622

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	622	4FA	OD25-H2SO4-PF1-N-6518-01	1 OF 2	0	CJ-M15W14-PR-A20-000109/145	1A/1A	
2	622	4FA	OD25-H2SO4-PF1-N-6518-02	2 OF 2	0	CJ-M15W14-PR-A20-000109/145	1A/1A	
3	622	5FA	OD25-H2SO4-PF1-N-6518-01	1 OF 1	0	CJ-M15W14-PR-A20-000109/145	1A/1A	
4	622	RFA	OD25-H2SO4-PF1-E-6518-01	1 OF 2	1	CJ-M15W14-PR-A20-000109/145	1A/1A	
5	622	RFA	OD25-H2SO4-PF1-E-6518-02	2 OF 2	1	CJ-M15W14-PR-A20-000109/145	1A/1A	



대외비

KEY PLAN (4F)

NOTE

1. 온천 시 CLOSE되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
악물종류에 따라 LINE 색상을 구분하여야 한다.

3. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는
DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로
이송된다.

4. PUMP VENDOR 공급.

5. 황산 LINE에 위치한 VALVE에 LOTO 적용.

6. AIC-W1054A/B 중 선택호전.

7. AIC-W1257, W1265 중 선택호전.

8. N-STRIPPER PACKAGE TRAIN A/B 중 선택호전.

9. 모든 황산 배관의 LEAK POINT에 대해서는
비산 방지 SHIELD 및 SENSING TAPE 설치할 것.

10. 황산 탱크 AREA에는 비산방지막을 설치한다.

11. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.

12. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa

13. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한진희	
승 APPROVED BY	박동영	
제 제 SK 에코플랜트		
공 PROJECT TITLE	M15X Ph4 WWT Project	
도 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (2/4)	
속 SCALE	NONE	
날 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145	1A

ITEM NO.	W14-CDP-H2SO4-4610 A/B/C	ITEM NO.	W14-CDP-H2SO4-4611 A/B/C	ITEM NO.	W14-CDP-H2SO4-4612 A/B/C
SERVICE	HF 1'st H2SO4 FEED PUMP	SERVICE	CIF 1'st H2SO4 FEED PUMP	SERVICE	AMMONIA ABSORBER TOWER H2SO4 FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA.	1/min x MPa	CAPA.	1/min x MPa	CAPA.	1/min.ea x MPa
MAT'L	1A (MAX. 11.2 x MAX 0.7 NOTE 13) PVDF/EPDM+PTFE/CERAMIC	MAT'L	1A (MAX. 11.2 x MAX 0.7 NOTE 14) PVDF/EPDM+PTFE/CERAMIC	MAT'L	1A (MAX. 11.2 x MAX 0.7 NOTE 14) PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75)	POWER kW	(0.75)	POWER kW	(0.75)



ST4			
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD



CLIENT



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CONTRACTOR 

PROJECT NAME

M15X Ph4 Project

COMMODITY	H2SO4
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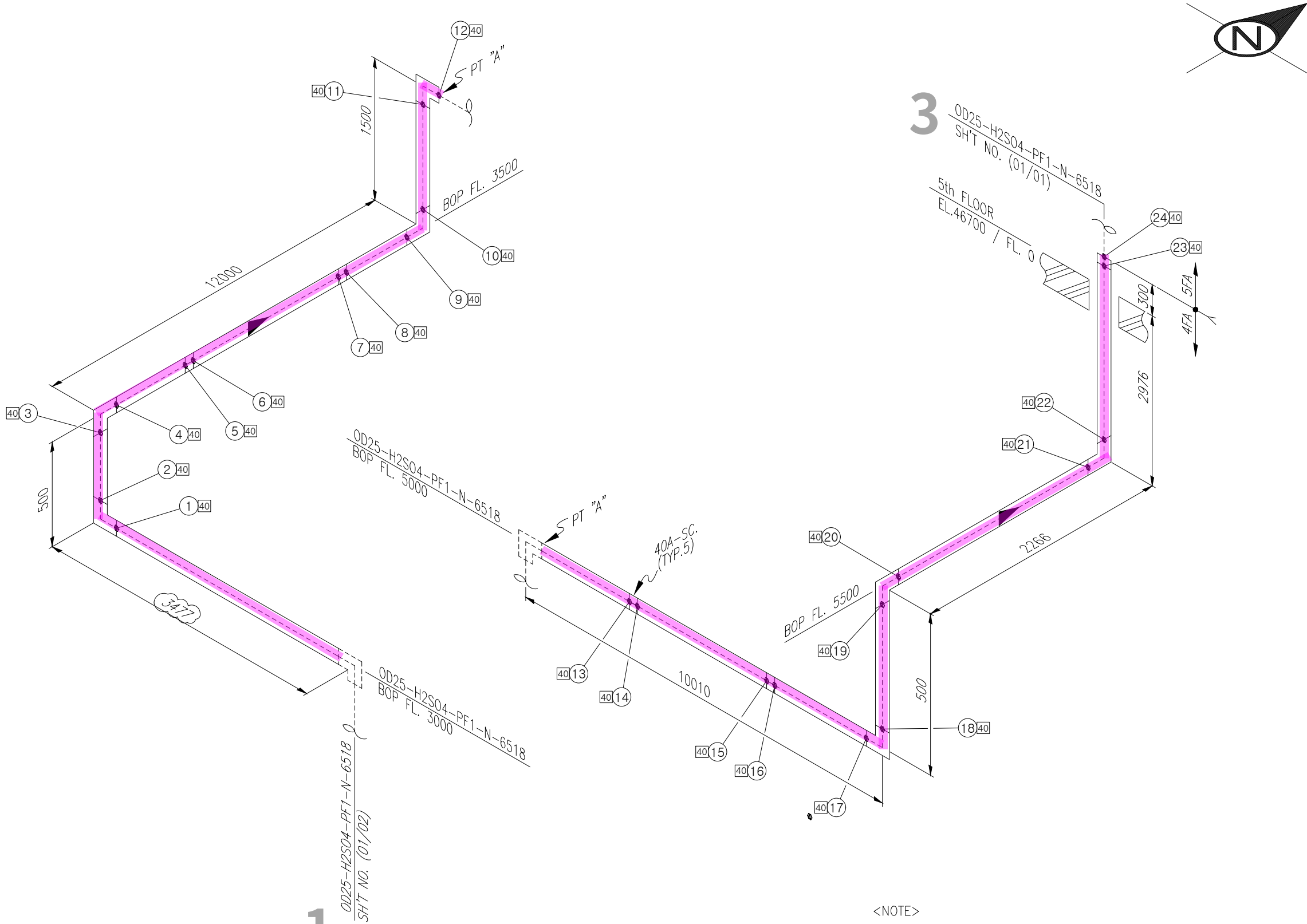
JOB NO.

22330D

LINE NO.	0D25-H2S04-PF1-N-6518
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SH'T NO.	
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<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR 	PROJECT NAME M15X Ph4 Project		
	THICK.	—	DESIGN	6.9	60	UNIT	4FA										
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A								COMMODITY	H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000145		△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD25-H2SO4-PF1-N-6518	SH'T NO. 02 / 02	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



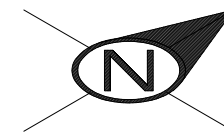
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1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	6.7	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000145										



1. (*) FIELD ADJUST



	CONTRACTOR 	PROJECT NAME M15X Ph4 Project	
영	COMMODITY H2SO4	JOB NO. 22330D	
영			
D	LINE NO. 0D25-H2SO4-PF1-E-6518	SH'T NO. 01 / 02	REV. 1

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INSULATION	CODE	E	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.9	60	UNIT RFA						
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000109		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음